Smart City Sim — Build Spec

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Shoreditch skyline

Pitch

A team of Claude agents that runs overnight and tells independent shops in Shoreditch when to open and how to light their store. It's a product for store owners. Under the hood it's a 3D sim where persona-driven agents live through 14-day runs, and the agents learn better **opening hours** and **lighting** run after run.

Pitch line: *"Your opening hours are a guess. Our agents live 14 days of Shoreditch overnight and tell you what to change."*

The business (decided)

Shops are the customer. Government is phase 2 and only ever buys anonymised area data. Everything in the build serves the store owner first.

Who's who

	Role	Pays for	Sees
Store owner	User and customer	Weekly report; monthly subscription per store	Only their own store
BID	Channel	Bulk seats for its members	Area totals, anonymised
Council	Phase 2 (the mic drop)	Area energy and footfall insight; street-lighting optimisation	Area totals, anonymised; never individual stores

Why a shop owner buys

- **Hours are a guess.** Owners set them by habit or copy neighbours; nobody can test "what if I closed at 3am on Fridays?" without risking real weeks of trade.
- **Energy is a big, visible cost.** Lighting an empty shop is money burned; lighting it at the wrong hour loses passers-by.
- **They have no analyst.** Chains have data teams; independents have nothing. We give them a team of agents that does the analysis overnight, every week.
- **Zero effort.** No hardware, no setup: we use public data on day one. Approve or reject in one tap.

Why us / what's hard to copy

- **Simulation, not dashboards.** Footfall and till analytics tell you what happened; we test what *would* happen before you change anything.
- **One sim per area serves every shop in it.** Each extra store in an area costs us very little, so margins grow as an area fills up.
- **Data advantage compounds.** Every approved or rejected suggestion and its real outcome makes next week's advice better for that area.
- **Long-running agents are the product, not a feature.** The overnight multi-agent run is what makes this affordable for a corner shop.

Government without breaking shop trust

The council goal (less energy, lower carbon, livelier high streets) mostly aligns with shops: **wasted light is wasted money and wasted carbon.** Four rules keep it clean:

1. **Data firewall:** store data is owned by the store; the council and competing stores never see it.
2. **No enforcement use:** never used for licensing or compliance. Said out loud in the pitch.
3. **Separate levers:** shops control their hours and lighting; the council controls street lighting. Neither optimises the other's.
4. **Independent advice:** each store's suggestions come from its own data plus public data, never coordinated across competitors (avoids competition-law risk).

Precedent: France already mandates shop-window lights off overnight. A voluntary, data-led alternative is an easy sell to councils.

Go-to-market

1. **Pilot one high street** (Shoreditch) through its BID: a handful of shops free for a month, measure real change.

2. **Convert to paid** per-store subscriptions; BID buys seats for members who want it.
3. **Repeat area by area** across UK high streets; each area = new map + local personas.
4. **Phase 2:** sell the anonymised area view and street-lighting optimisation to councils, using pilot results as proof.

Pitch arc

1. **~60%: one shop's story.** The dark shop next to the club, the agents running overnight, the report, the owner taps Approve, the replay shows a full shop.
2. **Zoom out:** "Now multiply by every shop on the high street." Area energy and footfall chart.
3. **Mic drop:** "Same engine, and the council gets carbon savings without touching anyone's business."
4. **Close:** business model, potential, responsible design.

Risks and answers

Risk	Answer
"Is the sim accurate?"	Calibration chart vs. Popular Times; every suggestion shows confidence; owner stays in control
Small shops won't pay	BID buys seats; free pilot; price anchored to £ saved per week
Shops fear the council sees their data	Data firewall + no-enforcement rule, written into terms
Competitors coordinating via us	Advice is per store, never cross-store
Council sales are slow	They're phase 2; BIDs bridge the gap

To validate (post-hackathon)

- [ ] Talk to 5 Shoreditch shop owners: would they pay, and how much?
- [ ] Talk to the Shoreditch BID: would they fund seats?
- [ ] Check competition-law and data terms with an adviser.

Why it fits Long Horizon: the value only shows up over many simulated days and many training iterations, not in a single step.

My take: strong idea. The demo moment is the before/after replay of the same 14 days. Protect that above everything else.

Product (store owners are the customer)

Problem: independent shops set opening hours by habit, miss demand (e.g. post-club crowds) and pay for lighting nobody sees.

Real output, per store, every week:

- 1–3 concrete changes, e.g. "Close 23:00 → 03:00 Fri–Sat", "Window lights off 20:00–22:00".
- Expected impact: visits, kWh, £ energy saved, with a confidence level.
- Evidence: the sim replay of that store, plus the data it's based on (cited).
- Owner approves or rejects; the agents learn from the decision next week.

Business model:

Item	Plan
Customer	Independent restaurants, cafes, shops, takeaways and bars, across the whole day (breakfast, lunch, dinner, late night)
Price	Monthly subscription per store (price point to validate)
Channel	Business Improvement District (BID) buys for its members; one area = one shared sim, so each extra store is cheap
Upsell	City view for BIDs and councils: footfall, peak energy, street lighting
Unit cost	Sim engine is rule-based (cheap); LLM spend only on agent reasoning, batched and cached. Show cost per store-week in the demo

City-level (government) view stays as an upsell, not the core product.

Potential: with more access and time

Each integration makes the sim more accurate and the product stickier.

More data access

- **Point-of-sale** (e.g. Square, SumUp, Lightspeed): real sales by hour, so suggestions are in £ revenue and the sim calibrates on each store's actual trade.
- **Smart lighting and plugs:** approved schedules applied automatically; measured kWh instead of estimates.

- **Footfall sensors and BID counters:** live street traffic to recalibrate weekly.
- **Events, weather and transport disruption feeds:** day-by-day hours, not just weekly.

More time

- **All-day verticals:** restaurants (lunch and dinner service windows), cafes (morning rush), retail (daytime), bars (late night). Same engine, different personas.
- **More levers:** staff rotas, promo and happy-hour timing, window displays.
- **More areas:** each new high street = a new map + local personas; sell area by area through BIDs, then to shopping centres, landlords and councils.
- **Data moat:** every approved or rejected suggestion and its real outcome trains the next week's agents.

Agents (the Long Horizon story)

Three layers of agents, each running on a longer time scale. The pitch line: *agents that live for days, and an agent that works for hours.*

Agent	Count	Lives for	Job	Long-horizon hook
Citizen	~300–500	One 14-day run	Walk, shop, eat, club per persona	Memory across days: "shop was shut Tue night" changes Wed route; habits form
Store owner	One per store	Across runs	Reads its own stats, proposes hours and lighting	Remembers what it already tried (the cache)
Analyst	1	Whole training session (hours)	Reads run logs, forms hypotheses, picks next experiment, decides when to stop	Runs unattended for ~10 iterations; keeps a persistent lab notebook

How to make the long-running part visible:

- **Run the analyst unattended before the demo** (e.g. overnight) and show the clock: "running 9h, 10 experiments".
- **Lab notebook in the History view:** one line per run, e.g. "Run 4: opened Kebab Co till 3am → +18% night visits". Short entries, no paragraphs.
- **Checkpoint and resume:** the analyst's state and notebook persist to disk, so a crash or restart picks up where it left off. Worth showing once.

- **Citizen memory within a run:** show one agent's 14-day diary as proof behaviour adapts over time.

Cost note: 500 LLM citizens per sim-tick is too expensive. Generate personas with an LLM once, run citizens on cheap rules, and call the LLM only for occasional decisions (e.g. once per sim-day). Spend the LLM budget on store and analyst agents.

Agent architecture on the Claude API

The analyst orchestrates a team of store agents; each Claude feature has a job, not a cameo.

Agent / part	Claude features	Why
Analyst (orchestrator, Opus)	Managed Agents session, multiagent orchestration, memory stores, define outcomes, session budgets	Runs for hours; notebook persists; stops on the threshold outcome; spend capped
Weekly re-run	Scheduled deployments, webhooks	Real product loop: runs every week, notifies the owner when the report is ready
Long context	Compaction, context editing	Hours of run logs without losing the thread
Store agents (team, Sonnet)	Batch processing, structured outputs, prompt caching	~25 proposals per iteration at batch price; shared city context cached; valid suggestion JSON every time
Sim as a tool	Tool use (`run_sim`, `get_kpis`, `get_store_stats`), code execution	Analyst runs experiments and analyses results itself (charts, significance)
Persona builder (Haiku)	Web search / web fetch, citations	Builds archetypes from public sources, every trait traceable
Owner report	Files API, citations	Downloadable weekly report with evidence

Model split: Opus for the analyst, Sonnet for store agents, Haiku for persona extraction. Cheap where volume is high, strong where judgement matters.

Scope: city vs store

One engine, two views: the store view is the product; the city view is the anonymised area view for BIDs and, later, councils. The sim is the same; only the objective and the zoom level change.

View	User	Optimizes for	Headline metric
City	Council / BID	Total energy vs. footfall	kWh saved, visits kept
Store	Shop owner	Own revenue vs. own energy	Visits per kWh, revenue

Lead the demo with the **store view** (concrete, relatable), then zoom out to the city to show the aggregate win. If time runs short, cut city-level optimization and keep it as a read-only stats overlay.

Levers

Keep it to two knobs per store so results are explainable.

Lever	Per store	Effect in sim
Opening hours	Open time, close time (30-min steps)	Captures flows, e.g. club spill-out at 2–4am
Lighting	Level per hour (off / low / high)	Raises attraction radius; costs kWh

Agents need a simple rule for lighting, e.g. a lit shop is more likely to be chosen at night, scaled by persona. Fixed anchors (clubs, stations, offices) are **not** optimized; they create the demand patterns.

Decision: double down on these two levers; deepen the energy model instead of adding levers.

Energy model

- **Baseline per store:** floor area (OSM footprint or EPC register) × typical kWh/m² for the store type.
- **Hourly carbon and price:** grid carbon intensity by hour, so the same light costs more at 6pm than at 2am.
- **City peak load:** sum all stores by hour into one demand curve; the city view shows the evening peak flattening.

Optional council lever: street lighting

Dim streets with no footfall; brighten the routes people take after the clubs. Gives the city view its own lever.

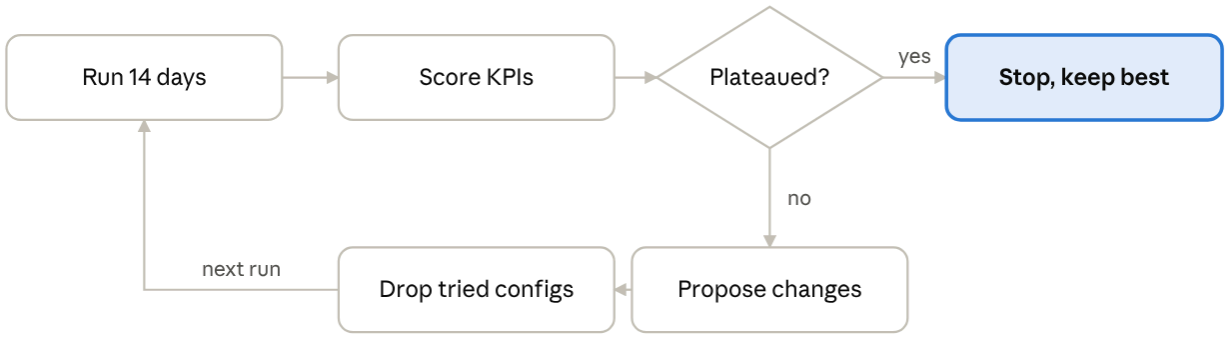
Out of scope

Store layouts and interiors (no public data; stores are boxes with a lit facade), plus pricing, stock and staffing.

Training loop

Each iteration = one 14-day run, then a suggestion step. Run serially; ~10 iterations before the demo.

Each run is scored; training stops once gains plateau



training loop · one run per iteration

- **Fixed seed:** compare runs with the same random seed and personas, so the delta comes from the changes, not noise.
- **Cache:** hash each store's config; never re-propose one already tried (stops flip-flopping back to old hours).
- **Threshold:** stop when KPI gain is below a set % for 2 runs in a row, or at 10 runs. Exact % is still open.
- **Objective:** one number, e.g. visits per kWh, or revenue minus energy cost.

Data (Shoreditch)

Scope to ~1 km² around Shoreditch High St: ~30–50 venues, ~300–500 agents. Use real data for the map and demand; use LLM-generated personas for behaviour.

Open boundary

The area is not a closed box: people walk in and out of Shoreditch, as in real life.

- **Entry/exit points:** streets that leave the map edge, plus stations. Agents spawn and despawn there.
- **Inflow by hour:** set arrival rates per edge and station from TfL and footfall data (commuters in the morning, club-goers at night).
- **Two visitor types in every training run:**
  - **Returning:** regulars (residents, office workers, repeat club-goers). Same identity across the 14 days, so they remember stores and adapt.
  - **One-off:** tourists and passers-by. Appear once, then leave for good.
- **Return rate as a KPI:** a good change should bring more people back, not just more first visits.
- **Fair comparison:** the fixed seed also fixes who arrives and who is a returner, so runs stay comparable.

Open question: returning vs one-off split (estimate from TfL and footfall data).

Use	Source	What it gives us
Map	OpenStreetMap	Buildings, venues, footprints, current `opening_hours`
Persona traits	Reddit (r/london, r/AskLondon, r/londonsocialclub) via PRAW	Habits and preferences, e.g. where people go after clubs
Persona traits	Reviews: Google Maps, Yelp, TripAdvisor, Time Out	What attracts or repels; visit times mentioned in text
Routines	ONS Time Use Survey	Hour-by-hour activity by age and work status
Routines	TfL open data	Station entry–exit counts incl. Night Tube
Routines	Resident Advisor, Dice, Eventbrite	Club and event times that drive late-night surges
Energy	Non-domestic EPC register	Floor area and energy rating per shop
Energy	Carbon Intensity API	Hourly grid carbon intensity for London

Use	Source	What it gives us
Calibration	Google Popular Times via BestTime, Outscraper or SerpAPI	Hourly busyness per venue (ground truth)
Calibration	London Datastore	High-street footfall

Pipeline: an LLM summarises Reddit and reviews into ~10 persona archetypes; Time Use and TfL give each a daily schedule; Popular Times checks the result.

Calibrate once: tune agents until simulated busyness roughly matches Popular Times for a few known venues. That one chart is your credibility slide.

Open question: check each source's terms before scraping; Google and Reddit both restrict it.

UI

The 3D city is the screen; everything else is a thin overlay. No paragraphs anywhere.

- **City view:** day/night cycle, agents as dots, stores glow by lighting level. Corner HUD: day counter, run #, 3 KPIs (visits, kWh, visits/kWh).
- **Store view:** click a store → side card with a 24h strip (open hours + lighting) and its KPIs vs. last run.
- **Suggestions:** diffs, not text, e.g. "Close 23:00 → 03:00" with a +/− delta. Buttons: *Apply to this store* | *Apply all*.
- **History:** a single line chart of the KPI across runs, with the analyst's one-line notebook entry per run; click a point to replay that run.
- Speed control (1×, 10×, 100×) so 14 days plays in ~1 minute.

Responsible design

- **Human in the loop:** agents only suggest; the owner approves every change. No automatic changes to real stores.
- **Honest uncertainty:** every suggestion shows confidence and "simulated, not guaranteed"; calibration chart shows how close the sim is to reality.
- **Privacy:** only public, aggregate data. No usernames or individual profiles stored; Reddit and reviews are reduced to anonymous archetypes.
- **Real-world constraints:** never suggest hours outside a store's licence or that ignore staff and neighbours (noise, late-night licensing). Hard rules checked in code, not left to the model.
- **Sustainability by default:** the objective rewards energy saved, not just footfall.

- **Spend and safety limits:** session budgets cap agent cost; every agent prompt, tool call and decision is logged and replayable.

Misuse cases and how we block them:

Misuse	Block
Council uses store data for licensing enforcement	Councils only get anonymised area totals; written into terms
A shop spies on a competitor	No store ever sees another store's data or suggestions
Competitors coordinate hours or prices through us	Advice is per store only; no cross-store recommendations
Tracking real individuals	Citizens are synthetic archetypes; no usernames or personal data kept
Landlord uses footfall to raise rents	Store-level data is owned by the store and shared only with its consent
Agents overspend or run away	Session budgets, hard stop, every action logged

Judging criteria map

Criterion	Weight	How we hit it	Priority action
Real output	20%	Working end-to-end run producing per-shop reports for real Shoreditch venues	Get a live link and saved run proof (logs, notebook, run files) early; real beats polished
Business viability	20%	Shops as customer, BID channel, council as phase 2; data gaps councils can't fill	Get 2–3 real quotes from Shoreditch shop owners during the hackathon
Agent design and architecture	15%	Orchestrator + team of store agents + sim as tool; planning, hand-offs, memory, checkpoint/resume	Show the notebook, the hand-off and a resume-after-crash live
Demo and pitch	15%	5-minute story: one shop, agents overnight, approve, zoom out to the council	Rehearse; have a one-page diagram for round 1 at the table

Criterion	Weight	How we hit it	Priority action
Demo clarity	15%	3D before/after of the same 14 days; one KPI chart	Pre-cached runs so nothing stalls live
Responsible design	15%	Human approval, licence rules in code, anonymous data only, honest confidence, spend caps, no enforcement use	Name the misuse cases and how we block them (below)

Demo plan

1. Let the analyst agent run ~10 iterations unattended before judging (e.g. overnight); cache every run so the demo is replay, not live compute. Open on its runtime clock and lab notebook.
2. Show run 1 at speed: club closes, crowd spills out, nearby shop is dark and shut.
3. Click that shop: suggestion card shows "open till 03:00, low light 01–03".
4. Jump to the final run: same night, shop is open and full. KPI line chart climbs then flattens at the threshold.
5. Open that shop's weekly report and approve the change (human in the loop). Then zoom out: city-wide kWh saved and cost per store-week.

Open questions

- [ ] Optimizer: simple per-store hill-climb, or an LLM proposing changes from run stats?
- [ ] Revenue model: fixed spend per visit by store type, or persona-based?
- [ ] Renderer: Three.js / R3F in browser, or a game engine?
- [ ] Which 3–4 anchor venues to model by name?